

Software Risk Assessment Approaches

Software risk assessment approaches encompass a range of techniques and methodologies used to identify, analyze, and mitigate potential risks in software development. Here are some key approaches:

- **Risk Identification:** Gathering input from stakeholders to identify potential risks related to technology, requirements, resources, and more.
- **Risk Analysis:** Evaluating the likelihood and impact of identified risks to prioritize them for action.
- **Risk Prioritization:** Ranking risks based on their probability of occurrence and potential impact, guiding the focus of risk management efforts.
- **Risk Evaluation:** Assessing the acceptability of risks and determining the need for further actions, such as risk mitigation strategies.
- **Risk Mitigation:** Implementing strategies to address identified risks, ensuring that they do not adversely affect the project's success.
- **Risk Monitoring and control:** Continuously tracking and updating the status of identified risks to ensure they are managed effectively.

These approaches are essential for ensuring that software projects are completed on time, within budget, and with the appropriate quality. By employing these methods, organizations can proactively manage risks and enhance the overall success of their software development efforts.

Software Risk Assessment Techniques

Here are the commonly used software risk assessment techniques (used in Software Project Management):

1. Brainstorming

- Team-based discussion to identify possible risks
- Encourages idea generation
- Useful in early project stages

2. Delphi Technique

Key Features

- Involves **multiple experts**
- **Anonymous responses**
- Conducted in **multiple rounds**
- Feedback is summarized after each round
- Reduces bias and group pressure

Example: A company developing a **cloud-based banking system** consults cybersecurity, database, and cloud experts separately to identify security risks. After several rounds, they agree that **data**

breach risk is the highest priority.

3. Risk Probability-Impact Matrix

The **Risk Probability-Impact Matrix** is a qualitative risk assessment tool used to evaluate and prioritize risks based on:

- **Probability (Likelihood)** of occurrence
- **Impact (Consequence)** on project objectives
- It helps categorize risks as **High, Medium, or Low** priority.

Risk impact/probability 5×5 Matrix

Impact ↓ / Probability →	1	2	3	4	5
5 (Very High)	M	H	H	VH	VH
4 (High)	M	M	H	H	VH
3 (Medium)	L	M	M	H	H

Impact ↓ / Probability →	1	2	3	4	5
2 (Low)	L	L	M	M	H
1 (Very Low)	L	L	L	M	M

Example (Software Project)

Risk: Frequent requirement changes

- Probability = 4 (High)
- Impact = 4 (High)

Risk Score = 16 → High Risk

Action: Implement change control process.

4. Risk Breakdown Structure (RBS)

The **Risk Breakdown Structure (RBS)** is a **hierarchical representation of risks**, organized by categories and subcategories, similar to a Work Breakdown Structure (WBS).

It helps systematically identify and classify project risks.

Typical RBS Categories (Software Project)

Level 1: Major Risk Categories

- 1. Technical Risks**
- 2. Management Risks**
- 3. Organizational Risks**
- 4. External Risks**

Level 2: Sub-Categories

1. Technical Risks

- Requirement instability
- Technology complexity
- Design issues
- Integration problems
- Performance failures

2. Management Risks

- Poor planning
- Cost overrun
- Schedule delays
- Resource allocation issues

3. Organizational Risks

- Lack of management support
- Staff turnover
- Communication gaps

4. External Risks

- Vendor delays
- Legal/regulatory changes
- Market changes
- Natural disasters

Sample RBS Structure (Hierarchical View)

1. Technical Risks

1.1 Requirement Risk

1.2 Design Risk

1.3 Technology Risk

2. Management Risks

2.1 Schedule Risk

2.2 Cost Risk

2.3 Resource Risk

3. External Risks

3.1 Vendor Risk

3.2 Legal Risk

3.3 Market Risk

Steps to Create RBS

- Identify major risk categories
- Break them into sub-categories
- Further divide into specific risks
- Use it during risk identification sessions

Example (Software Project)

For a **Healthcare Management System**:

- Technical Risk → Data security failure
- Management Risk → Budget overrun
- External Risk → Government compliance change

5. Checklist Analysis

Checklist Analysis is a risk identification technique that uses a **predefined list of common risks** (based on past projects and lessons learned) to systematically identify potential risks in a current project.

Example Scenario

In a **banking software project**, the team uses a checklist and

identifies:

- Requirement change risk
- Data security risk
- Vendor dependency risk

These are then analysed using a probability-impact matrix.

6. SWOT Analysis

Definition

SWOT Analysis is a strategic planning technique used to identify:

- **S** - Strengths
- **W** - Weaknesses
- **O** - Opportunities
- **T** - Threats

It helps in understanding internal and external factors affecting a software project.

Components

Internal Factors External Factors

Strengths Opportunities

Weaknesses Threats

Explanation of Each Component

Strengths (Internal - Positive)

- Skilled development team
- Strong technical expertise
- Good project management tools
- Strong financial support

Weaknesses (Internal - Negative)

- Lack of experience in new technology
- Poor communication
- Limited resources
- Inadequate testing process

Opportunities (External – Positive)

- Growing market demand
- New technology advancements
- Government support
- Expansion possibilities

Threats (External – Negative)

- High competition
- Regulatory changes
- Cybersecurity risks
- Vendor dependency

Example (Software Project)

Project: AI-Based Healthcare System

- **Strength:** Experienced AI team
- **Weakness:** Limited healthcare domain knowledge
- **Opportunity:** Increasing demand for digital health services
- **Threat:** Strict data privacy regulations

7. Failure Mode and Effects Analysis (FMEA)

Failure Mode and Effects Analysis (FMEA) is a **systematic risk assessment technique** used to identify potential failure points in a system, analyze their effects, and prioritize actions to reduce risk.

It is widely used for **technical risk evaluation** in software projects.

Key Terms

- **Failure Mode** – How something might fail
- **Effect** – Impact of the failure
- **Cause** – Reason for the failure
- **Detection** – Ability to detect the failure before it occurs

Risk Priority Number (RPN)

$$RPN = Severity \times Occurrence \times Detection$$

Where each factor is rated from **1 (Low) to 10 (High)**.

- **Severity (S)** - Impact of failure
- **Occurrence (O)** - Likelihood of failure
- **Detection (D)** - Likelihood of detecting before impact

Higher RPN → Higher priority risk

Steps in FMEA

- Identify system components or processes
- List possible failure modes
- Determine effects of each failure
- Assign S, O, D ratings
- Calculate RPN
- Prioritize corrective actions
- Recalculate RPN after mitigation

Example (Software Project)

Project: Online Banking System

Failure Mode Effect	S O D RPN
Login failure Users unable to access system	9 4 3 108
Data breach Loss of customer data	10 3 4 120
Slow response Poor user experience	6 5 5 150

Highest RPN = 150 → Priority action required

8. Decision Tree Analysis

Decision Tree Analysis is a **quantitative risk assessment technique** used to evaluate different decision alternatives under uncertainty.

It uses probabilities and Expected Monetary Value (EMV) to select the best option.

Expected Monetary Value (EMV)

- $EMV = Probability \times Impact(Cost or Benefit)$
- Choose the alternative with the **highest expected benefit** or

lowest expected cost.

Steps in Decision Tree Analysis

- Identify decision alternatives
- Determine possible outcomes
- Assign probabilities to each outcome
- Estimate cost/benefit for each outcome
- Calculate EMV
- Select the best alternative

Example (Software Project)

A company must choose between:

Option A: Develop In-House

- Success (0.7 probability) → Profit ₹10,00,000
- Failure (0.3 probability) → Loss ₹4,00,000

$$EMV(A) = (0.7 \times 10,00,000) + (0.3 \times -4,00,000)$$

$$EMV(A) = 7,00,000 - 1,20,000$$

$$EMV(A) = ₹5,80,000$$

Option B: Outsource

- Success (0.8 probability) → Profit ₹8,00,000
- Failure (0.2 probability) → Loss ₹2,00,000

$$EMV(B) = (0.8 \times 8,00,000) + (0.2 \times -2,00,000)$$

$$EMV(B) = 6,40,000 - 40,000$$

$$EMV(B) = ₹6,00,000$$

Choose **Option B (Outsource)** since EMV is higher.

9. Monte Carlo Simulation

- Uses statistical simulation
- Predicts project completion time or cost
- Suitable for large projects

10. Sensitivity Analysis

- Determines which risk has the greatest impact
- Shows how changes affect project outcomes

Quick Summary Table

Technique	Type	Used For
Brainstorming	Qualitative	Risk identification
Delphi	Qualitative	Expert consensus
Probability-Impact Matrix	Qualitative	Risk ranking
FMEA	Semi-Quantitative	Technical risk priority
Decision Tree	Quantitative	Cost/time decisions
Monte Carlo	Quantitative	Schedule/cost prediction